Fishing Magnet

PERMANENT MAGNET FOR JUNK RETRIEVAL

FISHING

GENERAL DESCRIPTION

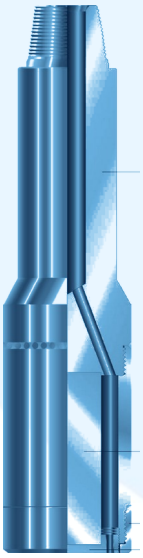
PERMANENT-MAGNET TOOL THAT RETRIEVES UNDRILLABLE OBJECTS WITH MAGNETIC ATTRACTION: BIT CONES, BEARINGS, SLIPS, MILL CUTTINGS, TONG PINS AND HAMMERS. IT PICKS UP SMALL, ODD-SHAPED ITEMS THAT INSIDE OR OUTSIDE CATCH TOOLS CANNOT HOLD. AVAILABLE FROM 1 IN. TO 20 IN. O.D., ON PIPE OR WIRELINE, WITH PULLS FROM 5 LB TO 5,000 LB DEPENDING ON SIZE. IT IS ALSO USED TO CLEAR THE HOLE BEFORE DIAMOND CORING.

TECHNICAL CHARACTERISTICS

O.D.	Hole size	Top pin	Approx. pull (lb)	Wt (lb)
1	1-1/4 to 2	5/8 11-N.C.	5 - 7	1
1-1/4	1-5/8 to 2-1/2	5/8 S.R.	8 - 10	1-1/2
1-1/2	2 to 2-3/4	5/8 S.R.	11 - 14	3-1/4
1-3/4	2-3/8 to 3-1/4	3/4 S.R.	15 - 20	5
2-1/4	2-3/4 to 3-5/8	3/4 S.R.	25 - 50	16
2-1/2	3 to 3-3/4	3/4 S.R.	50 - 85	18
3	3-1/4 to 4-1/8	2-3/8 Tbg	85 - 190	20
3-1/4	4 to 4-1/4	2-3/8 Tbg	85 - 190	25
3-1/2	4-1/4 to 4-1/2	2-3/8 API Reg.	150 - 250	27-1/2
4	4-1/2 to 5	2-3/8 API Reg.	175 - 250	43
4-1/2	5-1/8 to 5-1/2	2-7/8 API Reg.	250 - 320	67
5	5-5/8 to 6	2-7/8 API Reg.	320 - 385	80

O.D.	Hole size	Top pin	Approx. pull (lb)	Wt (lb)
5-1/2	6-1/8 to 6-1/2	3-1/2 API Reg.	385 - 425	95
5-3/4	6-1/8 to 6-1/2	3-1/2 API Reg.	385 - 425	106
6	6-5/8 to 7-1/2	3-1/2 API Reg.	425 - 500	120
7	7-5/8 to 8-1/2	4-1/2 API Reg.	550 - 700	162
8	8-5/8 to 9-3/4	4-1/2 API Reg.	700 - 850	205
9	9-7/8 to 11-5/8	4-1/2 API Reg.	850 - 1,000	335
10	10-1/2 to 11-7/8	6-5/8 API Reg.	1,000 - 1,125	390
10-1/2	11-3/4 to 13	6-5/8 API Reg.	1,125 - 1,260	431
11-1/2	12-1/4 to 14	6-5/8 API Reg.	1,260 - 1,550	512
14	15	6-5/8 API Reg.	1,700 - 2,100	770
16	17	6-5/8 API Reg.	3,200 - 4,000	1,200
19	20	13-3/8 V-4 thd	4,200 - 5,000	1,700

Dimensions in inches, weights in lb (complete assembly, approximate). S.R.: sucker rod pin; Tbg: tubing pin. Pulls assume the magnet face fully covers a flat surface; pick-up capacity drops considerably on smaller contact areas.



DESIGN FEATURES

- **Focused field:** flux is concentrated at the bottom pole plate; the outer case is not magnetised, so the tool works inside cased hole.
- **Circulation:** ports through the body exit inside the guide and keep the pole plate clean.
- **Guides:** flush (standard) for large or flat objects; lipped for 3 or more cones or leaning objects; mill guide for tong dies, 1 - 2 cones and safety runs before diamond coring.
- **Run on pipe or wireline:** wireline needs adaptors and gives no circulation.

OPERATING PARAMETERS

Approach	Stop 6 in. to 1 ft above the fish; circulate to wash
Contact	Reduce flow, lower onto fish, rotate lightly, stop pumps
Max. weight, rotating	5,000 lb
Max. weight, no rotation	10,000 lb
Proper charge	2,500 gauss or more

Excess weight or rotation can damage the pole plate and magnetic element. Never store two magnets face to face: the smaller loses charge. Clean ports after each run.

HOW TO ORDER

Please specify: 1. tool name and assembly number 2. O.D. 3. top connection if other than standard. Recommended accessories: 1 lipped guide and 1 mill guide.